Aman Khandelwal

Gameplay & Systems Programmer — Unreal Engine 5 / Unity

Jaipur, India (IST, UTC+5:30) · khandelwalaman5@gmail.com · +91 8619067110
GitHub: github.com/hitz482 · ArtStation: artstation.com/hitzsama · YouTube: @amankhandelwal8816 · LinkedIn

SUMMARY

Gameplay and systems programmer with 5+ years shipping multiplayer titles across PC, mobile, and VR in Unreal Engine 5 and Unity. Works from server-authoritative netcode and dedicated server deployment through to custom editor tooling. Shipped titles on Steam, Google Play, the App Store, and the Meta Quest Store; author of a solo-built UE5 auto-battler framework, ~13,800 lines of C++.

SELECTED ENGINEERING WORK

HITZ: Merge Tactics — UE5 Auto-Battler Framework (solo)

Unreal Engine 5.8 · ~13,800 lines of C++ across four modules · github.com/hitz482/HitzMergeTacticsCodeBase · docs

- Architected the framework as three reusable modules (`HMT_Core` 6.2k lines, `HMT_CoreEditor` 1.3k, `HMT_GASBridge` 0.3k) with the playable sample game kept in a separate fourth module (6.0k), so the framework lifts into an existing project by copying three source folders.
- Built server-authoritative combat: the server locks board snapshots, simulates, and replicates a compact event log that clients replay for animation — clients never compute outcomes, closing the cheat surface on match results.
- Designed the extension surface around four Blueprint-implementable interfaces (`IHMT_StatProvider`, `IHMT_TargetingStrategy`, `IHMT_TraitEffect`, `IHMT_AbilityExecutor`), so users ship a complete game without writing C++.
- Wrote a GAS bridge mapping the stat interface onto an existing `AttributeSet`, letting GAS-based projects swap the stat backend per-unit with zero changes to any other system.
- Built custom editor tooling — click-to-paint board layout editor, Unit Definition asset editor with live 3D preview and per-montage playback, and a save-time content validator.
- Implemented the full multiplayer round loop on standard UE actor replication: preparation/combat/results, recency-weighted pairing, ghost boards for odd lobbies, PvE rounds, spectating, late-join.
- Supports hex and square grid topologies from one board asset, with neighbour/distance/transform math decoupled from board state.

Ktrl Smash — Multiplayer Arena Brawler

Unity · Steam (Windows) + WebGL · Released Dec 2025 · Steam

- Shipped a free-to-play online PvP brawler, currently rated Mostly Positive on Steam.
- Built multiplayer on Photon Fusion; worked with the devops team to deploy dedicated servers on AWS.
- Diagnosed and fixed a session-long latency climb (20→60 ms Windows, 40→100 ms WebGL). Root cause was per-frame work accumulating on the main thread under WebGL's single-threaded runtime, compounded by over-frequent network updates and redundant replicated state. Audited coroutine usage across the codebase, cut update frequency and payload size, and removed unnecessary replicated variables — holding ping flat across full sessions.

The Cursed Hour — 4-Player Co-op Survival Horror

Unreal Engine 5 · Steam · Released Nov 2024 · ~18–23k estimated players, 129 reviews

- Pitched the original concept; owned core multiplayer systems through to ship.
- Built the ghost NPC, the lobby and session layer, and text and voice chat.
- Integrated both Epic Online Services and the Steam SDK in parallel against a planned cross-platform rollout, keeping the session layer backend-agnostic.

DinoMap Explorer — Location-Based AR Collection Game (client project)

Unity · Android & iOS · In development · Team Lead

- Lead developer on a location-based AR title where players discover and collect dinosaurs in the real world, with an educational layer.
- Built the AR capture flow, the in-game dinosaur encyclopedia, and NFT integration.
- Several systems built here became shared components reused across other studio titles.

EXPERIENCE

Senior Game Developer — Fiftyfive Gaming Studio Pvt Ltd

Oct 2021 – Present · Jaipur, India · Teams of 5–6 (2 devs, modeller, animator, UI designer)

- Gameplay, combat, and systems programming across shipped titles on PC (Steam), mobile (iOS/Android), and VR (Meta Quest).
- **Rift Riders** (UE, iOS/Android): introduced the concept, then built endless track generation, vehicle pawn, quest and task systems, and Firebase/Play Services integration.
- **Fight Hero** (Unity, iOS/Android): owned the mech combat loop, enemy waves, and upgrade/progression balancing; directed the art team on combat animation and gameplay assets.
- **Combat animation packs** (Fab, ArtStation): identified the product opportunity, directed the artist team, and built the UE integration layer — animation blueprints, state machines, root-motion and in-place variants — plus packaging and publishing. Generated ~$6,000 over two years and still earns passive revenue.
- Resolved a persistent iOS `bad_weak_ptr` crash caused by Firebase linking twice — Unity prebuilt libraries against EDM4U's Swift Package Manager resolution.
- **Tower of Demons** (Unity): added new game modes, cosmetics, resolution-scalable UI, and Firebase-backed player data.
- **Nutrients Sold Separately** (UE, Play Store): Epic game jam entry, team of 5, "As Above, So Below" theme.

Unreal Game Developer, Intern — Hyperreality Technologies

Jun 2021 – Oct 2021

- **Robo XR** (UE, Meta Quest) — sole developer on a robot build-and-battle VR title shipped to the Meta Quest Store. Built the robot assembly system, combat, enemy AI, and DLC support.

TECHNICAL

Languages: C++ (UE5 gameplay framework, replication, editor modules), C#
Multiplayer: Server-authoritative simulation, actor replication, dedicated servers (AWS), Photon Fusion, Epic Online Services, Steam SDK, matchmaking, voice/text chat
Unreal Engine 5: Blueprints, C++ modules, Gameplay Tags, GAS, custom editor tooling, data assets, MetaHuman, Cinematics
Unity: URP, Burst, WebGL and mobile builds, AR Foundation
AR/XR: Meta Quest, Oculus SDK, MapBox SDK, location-based AR
Backend: Firebase (Auth, Firestore, Cloud Functions), Google Play Services, AdMob
Profiling: RenderDoc, Unreal Insights / Session Frontend, GPU Visualizer
Tooling: Git, Perforce, Visual Studio, Xcode

EDUCATION

B.Tech, Computer Science (Cloud Computing specialization) — Poornima University, Jaipur · 2018–2022